Thermoreceptors are found on the first four tarsal segments on the legs. The receptor chordotonal is found on the anal cerci which respond to air or earth borne vibrations. The photoreceptors of the cockroach consists of a pair of compound eyes at the dorsal surface of the head. Each eye is formed of about 2000 simple eyes called the **ommatidia** (singular: *ommatidium*), through which the cockroach can receive several images of an object. This kind of vision is known as mosaic vision with more sensitivity but less resolution.

Arthropod eyes are called compound eyes because they are made up of repeating units, the ommatidia, each of which functions as a separate visual receptor.

What is the difference between compound eyes and simple eyes?

Why is mosaic vision with less resolution seen in cockroaches?

Excretory system

The Malpighian tubules are the main excretory organs of cockroach which help in eliminating the nitrogenous wastes from the body in the form of uric acid. Cockroach excretes uric acid, so it is **uricotelic**. In addition, fat body, nephrocytes, cuticle, and urecose glands are also excretory in function.

The malpighian tubules are thin, long, filamentous, yellow coloured structures attached at the junction of midgut and hindgut. These are about 100-150 in number and are present in 6-9 bundles. Each tubule is lined by glandular and ciliated cells and the waste is excreted out through the hindgut. The glandular cells of the malpighian tubules absorb water, salts, and nitrogenous wastes from the haemolymph and transfer them into the lumen of the tubules. The cells of the tubules reabsorb water and certain inorganic salts. By the contraction of the tubules nitrogenous waste is pushed into the ileum, where more water is reabsorbed. It moves into the rectum and almost solid uric acid is excreted along with the faecal matter.



Marcello Malpighi – described these tubules and called them vasa varicose. Meckel later called them Malpighian tubules.

Reproductive system

Cockroach is dioecious or unisexual. They have well developed reproductive organs. The male reproductive system consists of a **pair of testes, vasa deferentia**, **an ejaculatory duct**, **utricular gland**, **phallic gland** and the external genitalia. A pair of three lobed testes lies on the lateral side of the 4th and 6th abdominal segments. From each testis arises a thin vas deferens, which opens into the ejaculatory duct through the seminal vesicles. The ejaculatory duct is an elongated duct which opens out by the male gonopore lying ventral to the anus. A **utricular** or **mushroom shaped gland** is a large accessory reproductive gland, which opens into the anterior part of the ejaculatory duct. The seminal vesicles are present on the ventral surface of the ejaculatory duct. These sacs store the sperms in the form of bundles called **spermatophores**. The duct of **phallic** or **conglobate gland** also opens near the gonopore, whose function is uncertain. Surrounding the male genital opening are few chitinous and asymmetrical structures called **phallomeres or gonapophyses** which help in copulation.

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